

Explainable Multi-Agent Generative Recommendation System for Personalized Learning

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Abstract

This work presents an initial literature analysis focusing on four traditional approaches to e-learning recommendation systems. The objective is to examine their methodology, scientific contributions, application domains, and limitations. Through this comparative review, We highlight recurring issues such as a lack of adaptability, no personalized content generation, and no reliable explanations. These insights motivate the need for more advanced solutions that combine Agentic AI, Generative AI, Retrieval-Augmented Generation (RAG) and Explainable AI techniques. This analysis therefore establishes the foundation for the development of our proposed Explainable Multi-Agent Generative Recommendation System for Personalized Learning.

Key Words: Agentic AI, GenAI, XAI, multi-agent architectures, LLMs, RAG, RS, E-learning.

1 Introduction

Although traditional e-learning recommendation systems provide some level of personalization, they rely heavily on past data, use rigid models, and lack transparency. These limitations reduce their ability to generate personalized learning content and weaken user trust.

With the emergence of Generative AI, Large Language Models (LLMs) and agentic architectures in recent years, new opportunities have appeared to overcome these limitations.

Before introducing our multi-agent generative framework, we will focus on reviewing and comparing four representative traditional approaches in e-learning recommendation systems.

2 Selected Papers

Three representative papers were selected based on relevance to e-learning RS, methodological diversity, and their focus on traditional machine learning or deep learning approaches:

3 Selected Papers

Four representative papers were selected based on their relevance to e-learning recommendation systems, methodological diversity, and their focus on traditional machine learning, deep learning, or early LLM-based approaches.

- **Paper 1:** *A Systematic Review of Deep Learning-Based E-Learning Recommendation Systems* (2021)[1]. This paper offers an extensive survey of deep learning approaches applied to e-learning recommendation systems, covering architectures such as CNNs, RNNs, LSTMs, and deep belief networks. It analyzes their effectiveness in addressing challenges like feature extraction, sparsity reduction, and learner behavior modeling, while also identifying persistent issues such as cold-start¹, scalability, and lack of interpretability.

¹**Cold start:** when the system lacks enough data about new users or items to produce accurate recommendations.

- **Paper 2:** *Personalized E-Learning Recommendation System Based on a Hybrid Approach* (2023)[2]. This study proposes a hybrid recommendation framework that integrates content-based filtering² with collaborative filtering³ to enhance personalized course suggestions. The approach leverages learner profiles, course metadata, and similarity computations to improve recommendation accuracy, demonstrating how combining classical recommendation techniques can mitigate some limitations of standalone models.
- **Paper 3:** *Retrieval-Augmented Recommender System: Enhancing RS with LLMs* (2023)[3]. This paper investigates the feasibility of employing LLMs, specifically ChatGPT, as recommender systems. The authors propose a retrieval-augmented recommendation approach that combines traditional retrieval-based techniques with LLM-driven content generation. Through experiments on benchmark datasets such as MovieLens and Facebook Books, the study demonstrates that LLMs can perform competitive recommendations in zero-shot settings. Despite its promising results, the approach relies on prompt-based generation without explicit agent reasoning, planning mechanisms, or explainability frameworks, limiting transparency and controllability.

These papers serve as examples of traditional approaches enabling a structured comparison (in the next section) before proposing our advanced architecture.

4 Comparative Analysis

Table 1 below, presents a detailed comparison between the three papers based on: Domain, Approaches, Contributions and Limitaions.

Table 1: Comparison of Traditional and Emerging E-learning Recommendation Approaches

Aspect	Paper 1: DL Survey (2021)	Paper 2: Hybrid RS (2023)	Paper 3: LLM-based RS (2023)
Domain	Deep learning-based e-learning recommender systems evaluated on MOOCs and benchmark datasets.	Personalized course recommendation for e-learning platforms using learner profiles and course metadata.	E-learning and recommendation tasks enhanced using large language models (LLMs).
Approach	Survey of DL architectures (FNN, CNN, RNN/LSTM, DBN), hybrid CF/CB models, and ontology-based methods.	Hybrid recommendation combining content-based filtering and collaborative filtering through similarity measures.	Retrieval-augmented recommendation combining traditional retrieval methods with LLM-based prompt-driven generation.
Scientific Contributions	Provides a taxonomy of DL-based recommendation techniques and analyzes their effectiveness in modeling learner behavior.	Introduces a hybrid framework that improves recommendation accuracy through learner-course matching.	Shows that LLMs can deliver competitive zero-shot recommendations with enhanced semantic understanding and natural language interaction.
Limitations	Focuses mainly on existing models without proposing a unified implementation framework.	Relies on handcrafted features and similarity metrics, limiting scalability.	Lacks explicit reasoning, planning, and explainability mechanisms, reducing transparency and control.

²CB (Content-Based Filtering): method based on item features and user profiles.

³CF (Collaborative Filtering): method based on user-item interaction similarity.

5 Discussion: Limits of Traditional Approaches

The three papers show important progress in e-learning recommendation systems, but they are still limited to traditional prediction-focused models and do not address reasoning or content generation.

Lack of adaptability

Neither approach supports dynamic planning to create personalized learning paths. Systems are static and rely on historical data.

No personalized content generation

All approaches focus solely on recommending existing content. They do not integrate generative models capable of producing quizzes, summaries, or personalized learning materials.

Lack of reliable explanations

All approaches operate as black-box⁴ systems with no mechanisms to justify recommendations. These models reduce user trust.

6 Our Contribution

The previous analysis highlights clear gaps in traditional e-learning recommendation system approaches. To address these limitations, we propose our Explainable Multi-Agent Generative Recommendation System, which combines agentic architectures, Generative AI (GenAI), and Explainable AI (XAI) using technologies such as RAG and LLMs.

1. **Integration of Generative AI:** Dynamic educational content generation, including quizzes and recommending personalized learning materials adapted to the learner’s level.
 2. **Multi-agent architecture:** A coordinated system of specialized agents capable of reasoning, planning, communication, and task orchestration to support adaptive learning.
- Table 2 presents a description of the agents composing the proposed multi-agent architecture.

Table 2: Description of the Multi-Agent System

Agent	Description
Learner Profiling Agent	Analyzes learner performance data to determine knowledge levels, strengths, and weaknesses for each course. Produces a structured learner profile used by downstream agents.
Generative Content Agent	Generates adaptive quizzes and learning resources based on the learner’s knowledge level and weaknesses.
Recommendation and Ranking Agent	Selects the most suitable quiz per course by matching learner knowledge level and difficulty requirements. Enhances quizzes with correct answers and brief justifications.
Explainability Agent	Provides clear and encouraging explanations for recommended quizzes and resources. Justifies difficulty choices and supports learner understanding and motivation.

3. **Explainable AI techniques:** Transparent and human-interpretable explanations that justify recommendations and system decisions to learners.

⁴**Black box model:** a system whose internal decision-making is not interpretable.

Figure 1 summarizes the different layers that constitute our Explainable Multi-Agent Generative Recommendation System.

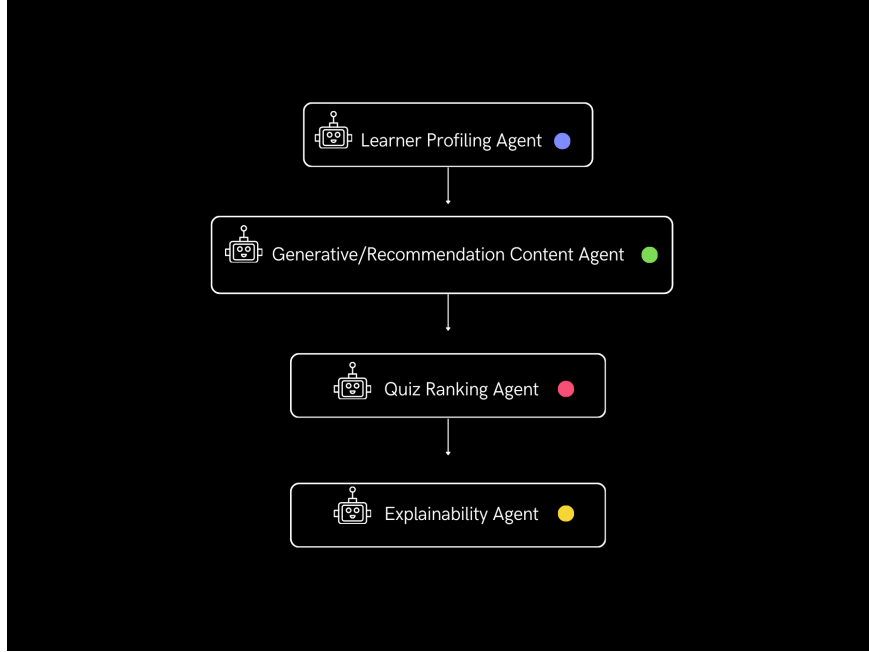


Figure 1: Our System’s Workflow

7 Technologies Used

Table 3 summarizes the main technologies used in the implementation of the proposed system.

Table 3: Technologies

Technology	Purpose
n8n	Workflow automation and orchestration of multi-agent interactions.
GPT-OSS-120B (via OpenRouter API)	Large Language Model used for learner profiling, content generation, recommendation, and explainability.
React	Frontend framework for building the interactive user interface.
JavaScript	Client-side logic and communication with backend web-hooks.

8 System Realization

This section describes the realization of the proposed system through an n8n-based multi-agent workflow.

8.1 Workflow Overview

1. The workflow begins with a **Set Node** that initializes learner data, including learner ID, past courses, and quiz scores, as shown below.

```
[
  {
    "learner_id": "L001",
    "past_courses": ["Python Basics", "Data Structures", "SQL"],
    "quiz_scores": [85, 53, 20]
  }
]
```

2. The data is then passed to the **Learner Profiling Agent**, which uses the GPT-OSS-120B model to analyze performance and determine the learner's knowledge level, recommended quiz difficulty, strengths, and weaknesses.

```
{
  "course_name": "SQL",
  "quiz_score": 20,
  "knowledge_level": "beginner",
  "recommended_quiz_difficulty": "easy"
}
```

Strengths and weaknesses are identified as follows:

```
"strengths": ["Python Basics"],
"weaknesses": ["SQL"]
```

3. **RAG Code Node** retrieves relevant learning materials from an internal course knowledge base based on the learner's identified course and knowledge level. These retrieved materials are used to enrich the content generation, ensuring that quizzes and recommendations are aligned with learner level and knowledge.

4. **Code Node** parses the profiling output and forwards it to the **Generative Content and Recommendation Agent**. This agent generates between 5 and 10 quizzes per course based on difficulty level and recommends learning resources (websites and YouTube videos) for weak areas.

5. After parsing, learner profile data and generated content are merged and passed to the **Ranking Agent**. This agent selects a single quiz per course based on the following criteria:

- Matching the learner's knowledge level
- Matching the recommended quiz difficulty
- Addressing learner weaknesses
- Focusing on core concepts

The selected quiz is enriched with correct answers and brief justifications, while preserving the recommended resources.

6. The output is then passed to the **Explainability Agent**, which generates concise and encouraging explanations for each recommendation:

```
{
  "course_name": "SQL",
  "explanation": "This quiz is set at a beginner level to strengthen  
your understanding of core SQL concepts.",
  "learning_tip": "Keep practicing basic queries to build
```

```
confidence."
}
```

7. Finally, the results from the ranking and explainability agents are merged and sent to the user interface.

Figure 2 shows our n8n workflow.

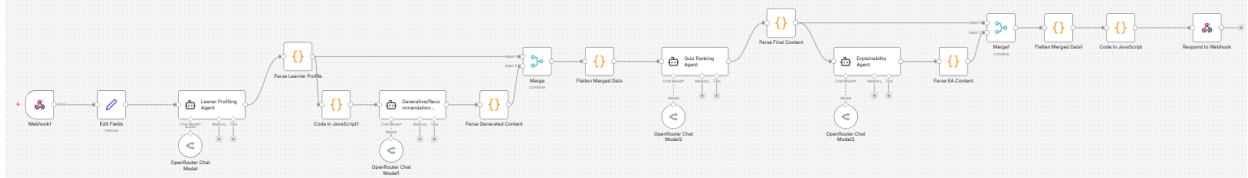


Figure 2: n8n Workflow

8.2 User Interface Integration

A **Webhook Node** is used to connect the n8n workflow to the frontend. When the webhook URL is triggered, the workflow executes and returns results to a React-based user interface developed using JavaScript. This enables real-time interaction between the learner and the recommendation system.

Figure 3 shows a partial view of the proposed user interface.

AI Learning Assistant

Enter your information to get personalized quiz recommendations.

Student ID

Past Courses (comma-separated)

Quiz Scores (comma-separated)

[Get Recommendations >](#)

SQL

Quiz: 1 of 3

Why this quiz?

This quiz was selected at a foundational level to strengthen your understanding of key SQL statements like SELECT and WHERE since these are essential basics. It will help you practice retrieving and filtering data effectively. If you find some concepts challenging, the provided website and video resources offer clear examples and explanations to support your learning.

Learning Tip: Keep building on your solid start with SQL fundamentals—they form the base for more advanced queries that you'll master soon!

Recommended Resources

Websites:

<https://www.w3schools.com/sql/>

Videos:

<https://www.youtube.com/watch?v=H0XV3zeQKqGY>

SQL Quiz

1 / 3
33% Score

1. Which keyword is used to retrieve data from a table?

☒ SELECT
 ☐ INSERT
 ☐ UPDATE
 ☐ DELETE

Explanation: SELECT is the SQL statement used to query and retrieve rows from a table.

Quiz Complete!

Student ID: Student002

Overall Score

11%

1 out of 9 correct

Figure 3: Part of the Proposed UI

9 Limitations

Despite its advantages, the system presents several limitations:

- The system currently relies on a single large language model (GPT-OSS-120B), which may limit the diversity and robustness of generated content.
- The system lack of quality Control mechanisms to verify the pedagogical accuracy or difficulty calibration of generated quizzes before they are presented to learners.
- The current implementation does not the learner history, which restricts the system’s ability to provide truly adaptive recommendations over time.
- The system lacks database integration for storing learner profiles, historical performance, generated content, and interaction.

10 Perspectives

The proposed system opens several directions for future research and improvement:

- Integrate multiple specialized language models combined with an improved retrieval-augmented generation to enhance generated content quality.
- Integrate fine-tuned LLMs specifically on educational content and pedagogical principles to improve the quality of generated content.
- Extend beyond text-based quizzes and links to generate multi-modal content(interactive visualizations, videos, simulation, etc)
- Build a complete system with account per learner and track each learner progress and performance to adapt the difficulty in generative content.

11 Conclusion

In this *draft paper*, we proposed an explainable multi-agent generative recommendation system for e-learning environments. The system combines agentic architectures, large language models, retrieval-augmented generation and explainable AI to generate personalized quizzes, recommend learning resources, and provide transparent explanations. Experimental realization demonstrates the feasibility of orchestrating specialized agents using an automated workflow.

References

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